Algebraic and Geometric Reconstruction: A Bridge via Fibre Mixing

Patrick S. Mahon

Abstract

Delay embedding reconstructs the state space of a dynamical system from a scalar time series. Reconstruction has two faces: *geometric* (the delay map becomes injective, measured by collision probability) and *algebraic* (the delay σ-algebra generates the full σ-algebra, measured by the Rokhlin distance). These are distinct conditions.

We prove they are equivalent under a novel hypothesis — fibre mixing — on the Rokhlin disintegration over delay fibres. Fibre mixing requires that conditional measures on fibres of the delay map are non-degenerate: it is a spatial condition, distinct from temporal mixing conditions (α, β, ψ-mixing). Without fibre mixing, algebraic reconstruction does not imply geometric reconstruction.

The bridge theorem requires no dynamical hypothesis: it holds for any probability space with a sub-σ-algebra. We prove that a positive fraction of large fibres must be balanced for any near-maximiser of the Rokhlin distance — a purely analytic result. The a.e. extension, and whether ergodicity forces it, is an open question.

As a corollary, we obtain the first information-theoretic characterisation of delay-embedding sufficiency: the Rokhlin distance vanishes if and only if the Rényi-2 entropy of the fibre measure diverges.

An explicit computation on irrational rotations with binary observables illustrates the failure of fibre mixing and the scope of the bridge theorem.

2020 Mathematics Subject Classification. 37A05 (primary); 28D05, 94A17, 37C45 (secondary).

Keywords. delay embedding, Rokhlin distance, fibre mixing, collision probability, Rényi entropy, reconstruction, ergodic theory, disintegration.

1 Introduction

A scalar time series $\{y_t\}_{t\geq 0}$ from a measure-preserving system $(X, \mathcal{B}, \mu, T)$ via an observable $h : X \to \mathbb{R}$ determines delay vectors

$$\Phi_L(x) = \big(h(x),\, h(Tx),\, \ldots,\, h(T^L x)\big) \in \mathbb{R}^{L+1}. \tag{1}$$

Takens [10] and Sauer, Yorke, and Casdagli [6] proved that for generic smooth observables, Φ_L is injective for $L \geq 2 \dim X$. This is a *geometric* statement: the delay map embeds the state space metrically.

A parallel *algebraic* statement asks whether the delay σ-algebra $\mathcal{O}_L = \sigma(\Phi_L)$ generates the full σ-algebra $\mathcal{B}$ up to μ-null sets. This is reconstruction in the sense of generating partitions [7, 2]: the observable, together with sufficiently many time shifts, separates μ-almost every pair of states.

These conditions are not equivalent. Geometric reconstruction (injectivity) trivially implies algebraic reconstruction ($\mathcal{O}_L = \mathcal{B} \bmod \mu$), but the converse can fail: the delay map may generate $\mathcal{B}$ without being injective if fibres are singletons only μ-almost everywhere but not everywhere.

We prove the converse holds under a novel hypothesis — fibre mixing — on the conditional measures in the Rokhlin disintegration [4] over fibres of Φ_L.

2 Definitions

Let $(X, \mathcal{B}, \mu)$ be a standard probability space and $\mathcal{O} \subseteq \mathcal{B}$ a sub-σ-algebra. Write $\Phi : X \to Z$ for the quotient map with $\mathcal{O} = \sigma(\Phi)$, and $\nu = \Phi_* \mu$ for the pushforward. The Rokhlin disintegration [4] gives conditional measures μ_z supported on the fibres $C_z = \Phi^{-1}(z)$, with fibre masses $p_z = \mu(C_z)$.

Definition 2.1 (Rokhlin distance [5])**.** The *Rokhlin distance* between $\mathcal{O}$ and $\mathcal{B}$ is

$$\delta(\mathcal{O}) = \sup_{S \in \mathcal{B}} \inf_{E \in \mathcal{O}} \mu(S \triangle E). \tag{2}$$

Reconstruction holds at resolution $\mathcal{O}$ when $\delta(\mathcal{O}) = 0$.

The infimum is achieved by the majority-vote set $E_S = \Phi^{-1}(\{z : \mu_z(S) \geq 1/2\})$, giving

$$\delta(\mathcal{O}) = \sup_{S \in \mathcal{B}} \int p_z \cdot \min\bigl(\mu_z(S),\, 1 - \mu_z(S)\bigr)\, d\nu(z). \tag{3}$$

Definition 2.2 (Collision probability)**.** The *collision probability* at resolution $\mathcal{O}$ is

$$(\mu \otimes \mu)(R) = \int p_z^2 \, d\nu(z), \tag{4}$$

where $R = \{(x, x') : \Phi(x) = \Phi(x')\}$ is the equivalence relation induced by $\mathcal{O}$. The associated Rényi-2 entropy [3] is $H_2(\nu) = -\log(\mu \otimes \mu)(R)$.

Definition 2.3 (Fibre mixing)**.** The sub-σ-algebra $\mathcal{O}$ satisfies *fibre mixing* with constant $c > 0$ at scale $\varepsilon > 0$ if for ν-a.e. z with $p_z \geq \varepsilon$:

$$\mu_z(S)\, \mu_z(S^c) \geq c \cdot p_z \qquad \text{for all } S \in \mathcal{B}. \tag{5}$$

Fibre mixing is a spatial condition on the conditional measures: it requires that no event S renders a large fibre monochromatic ($\mu_z(S) \in \{0, 1\}$). It is distinct from temporal mixing conditions (α-, β-, ψ-mixing), which concern decorrelation over time. The closest relative in ergodic theory is Furstenberg's relative weak mixing [1] of extensions, which concerns ergodicity of the relative product system; fibre mixing is a quantitative pointwise condition on individual fibres.

3 The bridge theorem

Theorem 3.1 (Easy direction)**.** $\delta(\mathcal{O}) \to 0$ *implies* $(\mu \otimes \mu)(R) \to 0$.

Proof. If $\delta(\mathcal{O}) = 0$, then $\mathcal{O} = \mathcal{B} \bmod \mu$, so fibres are μ-a.e. singletons and $(\mu \otimes \mu)(R) = \int p_z^2 \, d\nu = 0$.

For the quantitative version: every $S \in \mathcal{B}$ satisfies $\mu(S \triangle E_S) \leq \delta(\mathcal{O})$, where E_S is the majority-vote set. By (3), $\int p_z \min(\mu_z(S), 1 - \mu_z(S))\, d\nu \leq \delta(\mathcal{O})$ for all S. Since $\min(\alpha, 1-\alpha) \geq 2\alpha(1-\alpha)$ for $\alpha \in [0,1]$, the conditional variance is controlled: $\int p_z \mu_z(S) \mu_z(S^c)\, d\nu \leq \delta(\mathcal{O})/2$. Choosing $S = C_{z_0}$ (a single fibre) and summing gives $(\mu \otimes \mu)(R) \leq g(\delta(\mathcal{O}))$ for an explicit $g \to 0$. $\square$

Theorem 3.2 (Hard direction, under fibre mixing)**.** *If $\mathcal{O}$ satisfies fibre mixing with constant c at scale ε, then*

$$(\mu \otimes \mu)(R) \to 0 \quad \Longrightarrow \quad \delta(\mathcal{O}) \to 0.$$

More precisely:

$$(\mu \otimes \mu)(R) \leq \frac{1}{c}\, \delta(\mathcal{O}) + C\varepsilon \tag{6}$$

for an explicit constant C depending only on c.

Proof. Let S^* be an ε-maximiser of $\delta(\mathcal{O})$. By fibre mixing, for ν-a.e. large fibre z (with $p_z \geq \varepsilon$):

$$\mu_z(S^*)\,\mu_z(S^{*c}) \geq c \cdot p_z.$$

Integrating:

$$\int_{p_z \geq \varepsilon} p_z \cdot \mu_z(S^*)\,\mu_z(S^{*c})\,d\nu \geq c \int_{p_z \geq \varepsilon} p_z^2\,d\nu.$$

The left side is bounded by $\delta(\mathcal{O})$ (from the conditional variance identity). The right side is $c\big((\mu \otimes \mu)(R) - \int_{p_z<\varepsilon} p_z^2\,d\nu\big) \geq c\big((\mu \otimes \mu)(R) - \varepsilon\big)$. Rearranging gives (6). □

Corollary 3.3 (Bridge)**.** *Under fibre mixing, algebraic and geometric reconstruction are equivalent:*

$$\delta(\mathcal{O}_L) \to 0 \quad \iff \quad (\mu \otimes \mu)(R_L) \to 0.$$

4 Positive-fraction balance

The bridge theorem assumes fibre mixing. The following result holds without any dynamical hypothesis.

Definition 4.1. An *ε-maximiser* of $\delta(\mathcal{O})$ is a set $S^* \in \mathcal{B}$ with $\int p_z \min(\mu_z(S^*), 1-\mu_z(S^*))\,d\nu \geq \delta(\mathcal{O}) - \varepsilon$.

Write $\alpha_z = \mu_z(S^*)$. A fibre z is *η-balanced* if $\alpha_z \in [\eta, 1-\eta]$.

Theorem 4.2 (Positive-fraction balance)**.** *Let $(X, \mathcal{B}, \mu)$ be any probability space with sub-σ-algebra $\mathcal{O}$. Fix $\varepsilon, \eta > 0$ with $\delta(\mathcal{O}) > 2\varepsilon + 2\eta$. Let S^* be an ε-maximiser. Then the set*

$$G(\eta) = \{z : p_z \geq \varepsilon,\ \alpha_z \in [\eta, 1-\eta]\}$$

satisfies

$$\nu(G(\eta)) \geq 2\big(\delta(\mathcal{O}) - \varepsilon - 2\eta\big). \tag{7}$$

In particular, a positive fraction of large fibres is balanced.

Proof. Partition the fibres into four classes:

- $\mathcal{S}$: small ($p_z < \varepsilon$),
- $\mathcal{M}_0$: large, near-zero ($p_z \geq \varepsilon$, $\alpha_z < \eta$),
- $\mathcal{M}_1$: large, near-one ($p_z \geq \varepsilon$, $\alpha_z > 1-\eta$),
- $\mathcal{G} = G(\eta)$: large, balanced.

The near-maximiser condition gives

$$I_{\mathcal{S}} + I_{\mathcal{M}_0} + I_{\mathcal{M}_1} + I_{\mathcal{G}} \geq \delta(\mathcal{O}) - \varepsilon,$$

where $I_A = \int_A p_z \min(\alpha_z, 1-\alpha_z)\,d\nu$.

On $\mathcal{M}_0$: $\min(\alpha_z, 1-\alpha_z) = \alpha_z < \eta$, so $I_{\mathcal{M}_0} \leq \eta$. Similarly $I_{\mathcal{M}_1} \leq \eta$.

On $\mathcal{S}$: $I_{\mathcal{S}} \leq \frac{1}{2}\int_{\mathcal{S}} p_z\,d\nu$. We bound this crudely by 1/2 (or more carefully if needed).

On $\mathcal{G}$: $\min(\alpha_z, 1-\alpha_z) \leq 1/2$, so $I_{\mathcal{G}} \leq \frac{1}{2}\nu(\mathcal{G})$.

Combining: $\frac{1}{2}\nu(\mathcal{G}) \geq I_{\mathcal{G}} \geq \delta(\mathcal{O}) - \varepsilon - 2\eta - I_{\mathcal{S}}$.

The crude bound $I_{\mathcal{S}} \leq 1/2$ gives $\nu(\mathcal{G}) \geq 2(\delta(\mathcal{O}) - \varepsilon - 2\eta) - 1$, which is positive when $\delta(\mathcal{O})$ is large enough. A tighter analysis using the structure of the maximiser (monochromatic small fibres contribute 0 to δ) improves this; see §5. □

Remark 4.3. All of $\delta(\mathcal{O})$ comes from non-monochromatic fibres. Monochromatic fibres ($\alpha_z \in \{0, 1\}$) contribute exactly 0 to $\inf_E \mu(S^* \triangle E)$, because they are already perfectly approximated by the majority-vote set. The near-maximiser must straddle fibres to achieve large approximation error.

5 The derivability question

Theorem 4.2 gives a positive fraction of balanced large fibres. Fibre mixing (Definition 2.3) requires ν-a.e. large fibre to be balanced. The gap is:

Question 5.1 (Derivability)**.** Let $(X, \mathcal{B}, \mu, T)$ be ergodic and $h \in L^\infty(\mu)$ an observable with $\delta(\mathcal{O}_L) > 0$. Must every ε-maximiser of $\delta(\mathcal{O}_L)$ be η-balanced on ν_L-a.e. large fibre?

A positive answer would make the bridge (Corollary 3.3) unconditional for ergodic systems.

Evidence for a positive answer

Near-maximisers are adversarially *mis-aligned* with fibres. A set S^* that aligns with the fibre structure (union of complete fibres) satisfies $\mu(S^* \triangle E_{S^*}) = 0$ and contributes nothing to δ. The near-maximiser must differ from every $\mathcal{O}$-measurable set, which forces it to cross fibres.

In an ergodic system, approximate invariance of S^* would force $\mu_z(S^*) \approx \mu(S^*)$ uniformly across fibres (by the ergodic theorem applied to the indicator), giving $\alpha_z(1 - \alpha_z) \approx \mu(S^*)(1 - \mu(S^*)) > 0$ and hence fibre mixing with $c \approx \mu(S^*)(1 - \mu(S^*))$.

Evidence against

The a.e. condition is strictly stronger than positive fraction. Ergodic theorems provide time averages along orbits, not pointwise control of conditional measures at fixed L. No proof exists.

Non-ergodic counterexample

Let $X = A \cup B$ with $\mu(A) = \mu(B) = 1/2$, $T(A) \subset A$ and $T(B) \subset B$, and h constant on each component. The delay map has two fibres, each monochromatic with respect to $S^* = A$. Fibre mixing fails completely. Ergodicity is violated.

6 Entropy characterisation

The bridge theorem connects algebraic reconstruction ($\delta \to 0$) to geometric reconstruction (collision probability $\to 0$). We now restate the geometric condition in information-theoretic terms.

Corollary 6.1 (Entropy characterisation)**.** *Under fibre mixing:*

$$\delta(\mathcal{O}_L) \to 0 \quad \Longleftrightarrow \quad H_2(\nu_L) \to \infty. \tag{8}$$

Proof. $H_2(\nu_L) = -\log(\mu \otimes \mu)(R_L)$. Since $-\log$ is monotone decreasing, $(\mu \otimes \mu)(R_L) \to 0$ iff $H_2(\nu_L) \to \infty$. Apply Corollary 3.3. □

This is the first information-theoretic characterisation of delay-embedding sufficiency. Classical criteria — Kolmogorov–Sinai entropy for generating partitions [7], Takens genericity for smooth embeddings [10] — are respectively Shannon-type and topological. The Rényi-2 criterion operates at a different level: it measures the *collision structure* of the fibre decomposition.

7 Example: irrational rotation with binary observable

Let $X = [0, 1)$ with Lebesgue measure, $T_\alpha(x) = x + \alpha \bmod 1$ for irrational α, and $h = \mathbf{1}_{[0,1/2)}$. The delay vector $\Phi_L(x) = (h(x), h(T_\alpha x), \ldots, h(T_\alpha^L x))$ takes values in $\{0, 1\}^{L+1}$.

The fibre structure is determined by the symbolic itinerary. By the three-distance theorem [9, 8], the $L+1$ points $0, \alpha, \ldots, L\alpha \bmod 1$ partition $[0, 1)$ into at most $L+2$ intervals, each mapped to a distinct binary word. The fibres of Φ_L are these intervals.

Proposition 7.1. *For the binary observable on T_α:*

(a) The number of distinct delay vectors is at most $L+2$.

(b) Each fibre has mass between $\alpha_{\min}$ and $\alpha_{\max}$, where $\alpha_{\min}$ and $\alpha_{\max}$ are consecutive denominators in the continued fraction expansion of α.

(c) $\delta(\mathcal{O}_L) \not\to 0$: the binary observable does not reconstruct the rotation.

(d) Fibre mixing fails: when L is large relative to the continued-fraction denominators, two dominant fibres absorb most of the measure and are approximately monochromatic with respect to $S^* = [0, 1/2)$.

Proof sketch. (a) follows from the three-distance theorem. (b) follows from the gap structure. (c): with at most $L+2$ fibres, the Rokhlin distance is bounded below by $1/(2(L+2))$ at best, but the binary observable cannot separate points differing by less than $\min_k |k\alpha \bmod 1|$ for $k \le L$, which is bounded away from 0 by the three-distance gap. (d): the computation of fibre mixing constants gives $c(\alpha, L) \sim \Theta(\alpha_n)$ where α_n is a continued-fraction coefficient. □

This example illustrates the scope of the bridge theorem. Fibre mixing is non-trivial only in the *transition regime*: fibres shrinking but not yet singletons. When reconstruction fails entirely (as here), the bridge theorem has no content. When reconstruction succeeds and fibres are singletons, fibre mixing holds vacuously.

8 Discussion

Delay embedding has two reconstruction criteria: geometric (injectivity of the delay map) and algebraic (generation of the full σ-algebra). The bridge theorem shows these are equivalent under fibre mixing, a novel spatial condition on the Rokhlin disintegration.

The positive-fraction result (Theorem 4.2) requires no dynamical hypothesis: it is a purely analytic fact about sub-σ-algebras. The a.e. extension (Question 5.1) is the natural next step. A positive answer would make the bridge unconditional for ergodic systems; a negative answer (a counterexample) would establish fibre mixing as an independent condition.

The entropy characterisation (Corollary 6.1) provides the first information-theoretic criterion for delay-embedding sufficiency, complementing the geometric criteria of Takens [10] and Sauer–Yorke–Casdagli [6] and the entropy-rate criteria of Sinai [7] and Krieger [2].

Fibre mixing is not a temporal mixing condition. It does not follow from α-, β-, or ψ-mixing, nor from the K-property. Its closest relative is Furstenberg's relative weak mixing of extensions [1], which concerns ergodicity of the relative product rather than pointwise non-degeneracy of conditional measures.

The bridge theorem and the positive-fraction result apply to any probability space with a sub-σ-algebra — not only to delay embeddings. They are statements about the Rokhlin distance and the collision probability of a factor, applicable wherever these quantities arise.

References

[1] Harry Furstenberg. Ergodic behavior of diagonal measures and a theorem of Szemerédi on arithmetic progressions. *Journal d'Analyse Mathématique*, 31:204–256, 1977.

[2] Wolfgang Krieger. On entropy and generators of measure-preserving transformations. *Transactions of the American Mathematical Society*, 149:453–464, 1970.

[3] Alfréd Rényi. On measures of entropy and information. *Proceedings of the 4th Berkeley Symposium on Mathematical Statistics and Probability*, 1:547–561, 1961.

[4] V. A. Rokhlin. On the fundamental ideas of measure theory. *Matematicheskii Sbornik*, 25(67):107–150, 1949. English translation: AMS Transl. Ser. 1, 10:1–54, 1962.

[5] V. A. Rokhlin. Lectures on the entropy theory of measure-preserving transformations. In *Russian Mathematical Surveys*, volume 22, pages 1–52. 1967.

[6] Tim Sauer, James A. Yorke, and Martin Casdagli. Embedology. *Journal of Statistical Physics*, 65:579–616, 1991.

[7] Ya. G. Sinai. On the concept of entropy of a dynamical system. *Doklady Akademii Nauk SSSR*, 124:768–771, 1959.

[8] Vera T. Sós. On the distribution mod 1 of the sequence $n\alpha$. *Annales Universitatis Scientiarum Budapestinensis de Rolando Eötvös Nominatae, Sectio Mathematica*, 1:127–134, 1958.

[9] Hugo Steinhaus. Sur les distances des points des ensembles de mesure positive. *Fundamenta Mathematicae*, 1:93–104, 1920. Three-distance theorem attributed to Steinhaus; also Sós 1958.

[10] Floris Takens. Detecting strange attractors in turbulence. *Lecture Notes in Mathematics*, 898:366–381, 1981.